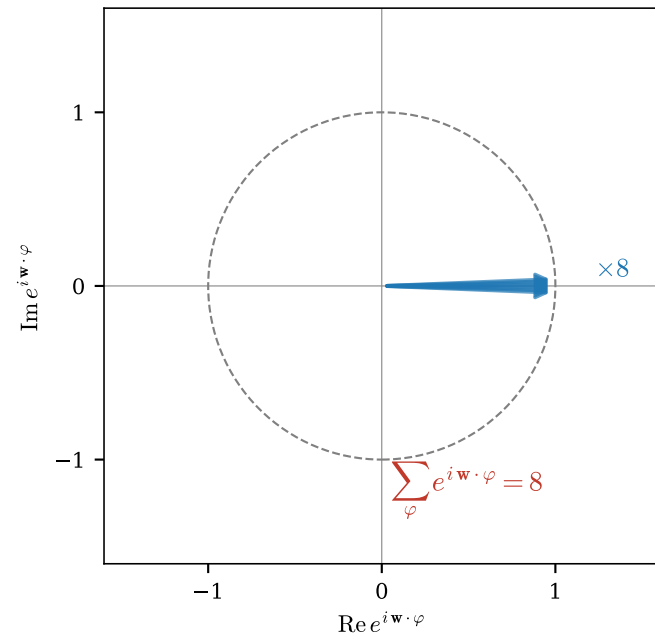
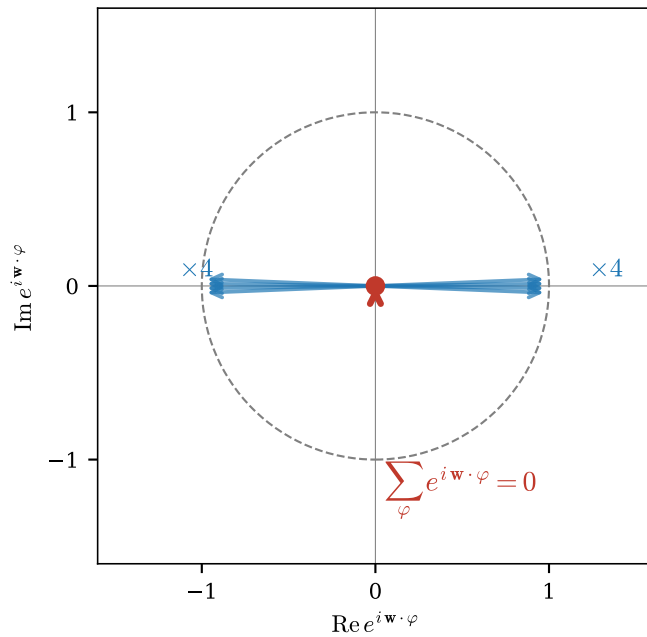


Phasor sum over the 8 corners $\varphi \in \{0, \pi\}^3$: $\frac{1}{8} \sum_{\varphi} e^{i\mathbf{w} \cdot \varphi} = \delta_{\mathbf{w} \bmod 2, 0}$

(a) $\mathbf{w} = (0, 0, 0)$: $e^{i\mathbf{w} \cdot \varphi} = 1$



(b) $\mathbf{w} = (1, 0, 0)$: $e^{i\varphi_x} \in \{+1, -1\}$



(c) $\mathbf{w} = (1, 1, 0)$: $e^{i(\varphi_x + \varphi_y)} \in \{\pm 1\}$

